

Detection of Differential Gene Flow from Patterns of Quantitative Variation

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# *Detection of Differential Gene Flow from Patterns of Quantitative Variation*

JOHN H. RELETHFORD[1] AND JOHN BLANGERO[2]

*Abstract* A major goal in anthropological genetics is the assessment of the effects of different microevolutionary forces. Harpending and Ward (1982) developed a model that aids in this effort by comparing observed and expected heterozygosity within populations in a local region. The expected heterozygosity within a population is a function of the total heterozygosity of the entire region and the distance of the population from the regional mean centroid of allele frequencies. Greater than average gene flow from an external source will result in a population having greater heterozygosity than expected. Less than average gene flow from an external source will result in a population having less heterozygosity than expected. We extend the Harpending-Ward model to quantitative traits using an equal and additive effects model of inheritance. Here the additive genetic variance within a population is directly proportional to heterozygosity, and its expectation is directly proportional to the genetic distance from the centroid. Under certain assumptions the expectations for phenotypic variances are similar. Observed and expected genetic or phenotypic variance can then be compared to assess the effects of differential external gene flow. When the additive genetic covariance matrix or heritabilities are not known, the phenotypic covariance matrix can be used to provide a conservative application of the model. In addition, we develop new methods for estimation of the genetic relationship matrix ($R$) from quantitative traits. We apply these models to two data sets: (1) six principal components derived from twenty dermatoglyphic ridge count measures for nine villages in Nepal and (2) ten anthropometric measurements for seven isolated populations in western Ireland. In both cases both the univariate and multivariate analyses provide results that can be directly interpreted in terms of historically known patterns of gene flow.

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A variety of methods have been developed to study the genetic structure of human populations. Relethford and Lees (1982) classify these methods into two groups: (1) model-free methods and (2) model-bound methods. Model-free methods rely on statistical analogies to specific population structure models, whereas model-bound methods are concerned with estimation of parameters from direct application of a theoretical model to data. An example of these two approaches is the relationship between genetic similarity and geographic distance. One can look at the overall correlation between genetic distance and geographic distance, or one can fit data to a specific model (e.g., isolation by distance).

Most model-fitting procedures rely on some measure of "goodness of fit" to evaluate the overall appropriateness of a specific model to a given set of data. In the past much of the work using model-bound methods consisted of estimation of relevant parameters and computation of some measure of fit. Such procedures do not provide information about how and why the model is (or is not) appropriate. Some studies, however, went beyond the assessment of degree of fit. The residuals from a model were examined to determine whether there was any systematic pattern of deviation from the basic model. For example, both Workman et al. (1976) and Relethford et al. (1981) looked at patterns of residuals from the isolation by distance model. In such cases the deviation of data from an expected model and the specific pattern of deviation may tell us more about genetic structure than simple estimation of parameters and goodness of fit.

Harpending and Ward (1982) presented a simple model that is useful in examining deviations from an expected pattern of genetic variation. Their model involves computation of the expected relationship between a population's heterozygosity and its genetic distance to a regional mean. Expected heterozygosity can then be compared to observed heterozygosity. The direction and magnitude of the difference between expected and observed heterozygosity provides information about differential gene flow from outside the region of analysis. Although many migration matrix methods make the assumption that the source of all external gene flow is from a homogeneous "outside world," the Harpending-Ward model aids in detecting which populations (if any) violate this assumption.

The Harpending-Ward model was developed for use with allele frequency data. In addition to a number of applications in their original paper, the method has been used successfully in other studies [e.g., Devor et al. (1984) and Relethford (1985)]. Our purpose here is to extend the Harpending-Ward model to the analysis of quantitative traits. We develop both univariate and multivariate extensions of the Harpending-Ward model. We also examine how robust our model is with respect

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to estimation of quantitative genetic parameters. We then apply our model to dermatoglyphic data from nine Nepalese populations and to anthropometric data from seven Irish populations.

## The Harpending-Ward Model

Given $n$ loci with two alleles at each locus, the Harpending-Ward model states that the expected heterozygosity of population $i$ [$E(H_i)$] is a function of the heterozygosity of the total region ($H_t$) and the genetic distance between population $i$ and the regional centroid ($r_{ii}$). That is,

$$E(H_i) = H_t(1 - r_{ii}). \tag{1}$$

This equation provides an expected linear regression line of heterozygosity on genetic distance from the centroid with intercept $H_t$ and slope $-H_t$. For two alleles at each locus the heterozygosity of the total region is computed under the assumption of complete panmixia as

$$H_t = \sum 2\bar{p}_k\bar{q}_k/n, \tag{2}$$

where $\bar{p}_k$ and $\bar{q}_k(= 1 - \bar{p}_k)$ are the weighted mean allele frequencies for locus $k$ and summation is over all $n$ loci. The mean allele frequencies are computed as

$$\begin{aligned} \bar{p}_k &= \sum w_i p_{ik}, \\ \bar{q}_k &= 1 - \bar{p}_k, \end{aligned} \tag{3}$$

where $w_i$ is the ratio of the census size of population $i$ to the total census size over all groups, $p_{ik}$ is the frequency of one allele at locus $k$ in population $i$, and summation is over all groups. It is important to note that these weights are based on relative census sizes, not on relative sample sizes.

The genetic distance of a population to the regional centroid is computed as the diagonal of the $R$ matrix of scaled variances and covariances about the regional mean allele frequencies (Harpending and Jenkins 1973; Workman et al. 1973). For each allele the elements of the $R$ matrix for populations $i$ and $j$ are computed as

$$r_{ij} = (p_i - \bar{p})(p_j - \bar{p})/\bar{p}\,(1 - \bar{p}). \tag{4}$$

The overall $R$ matrix is then averaged over all alleles. The $R$ matrix provides an estimate of genetic kinship relative to the contemporary

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region. That is, it measures deviations from the contemporary mean allele frequencies.

The observed heterozygosity of population $i$ is computed as

$$H_i = \sum 2p_{ik}q_{ik}/n, \tag{5}$$

where summation is over all $n$ loci. Under the assumption that all populations experience the same amount of gene flow from the same source (a homogeneous "outside world"), the expected heterozygosity and observed heterozygosity for population $i$ will be the same. If either the rate or source of external gene flow is different among populations, then the expected relationship will not hold for all populations. In particular, Harpending and Ward show that populations having greater than average external gene flow will have observed heterozygosities greater than expected. Comparison of expected heterozygosity (Eq. 1) with observed heterozygosity (Eq. 5) allows assessment of which populations, if any, have experiences greater than average external gene flow [$H_i > E(H_i)$] or less than average external gene flow [$H_i < E(H_i)$]. In studies for which independent information about external gene flow exists, the utility of the Harpending-Ward model can be tested. Given the method's success in previous studies, we believe that it can be used to detect differential external gene flow in studies for which independent knowledge of migration is not available.

Application of the Harpending-Ward model allows detection of differential admixture but may not necessarily reveal differences resulting from genetic drift, as has been claimed elsewhere (Devor et al. 1984). Populations lying below the expected regression line are considered more isolated from external gene flow but may still experience considerable gene flow from other populations within the region.

## Univariate Extension to Quantitative Traits

The Harpending-Ward model can be extended for use with polygenic quantitative traits. We present two cases, each with its attendant assumptions. First, we directly extend the Harpending-Ward model for use with additive genetic variances. Second, under more restrictive assumptions we consider how phenotypic variances can be predicted by the model. Both cases are considered in terms of univariate and multivariate extensions.

Direct extension of the Harpending-Ward model to quantitative traits requires the assumption that the additive genetic variance of a trait is proportional to heterozygosity. We assume additive effects over

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multiple loci, each with two alleles. Let $\alpha_k$, 0, and $-\alpha_k$ represent the genotypic values of the three genotypes at locus $k$. We further assume a model of equal effects, such that $\alpha_k = \alpha$ for all loci. According to traditional quantitative genetics models (Falconer 1981), the additive genetic variance within the $i$th subdivision is

$$\sigma^2_{Gi} = \sum 2\alpha^2 p_{ik}(1 - p_{ik}), \tag{6}$$

where summation is over all loci. The additive genetic variance for the total region assuming panmixia is

$$\sigma^2_{Gt} = \sum 2\alpha^2 \bar{p}_k(1 - \bar{p}_k). \tag{7}$$

The heterozygosity in subdivision $i$ can now be expressed as a function of the additive genetic variance:

$$H_i = \sigma^2_{Gi}/n\alpha^2. \tag{8}$$

Likewise, the heterozygosity in the total region under panmixia can be expressed:

$$H_t = \sigma^2_{Gt}/n\alpha^2. \tag{9}$$

Substitution of Eqs. (8) and (9) into Eq. (1) gives

$$E(\sigma^2_{Gi}) = \sigma^2_{Gt}(1 - r_{ii}). \tag{10}$$

The additive genetic variance in the panmictic region ($\sigma^2_{Gt}$) cannot be directly observed. This variance is *not* the same as the total additive genetic variance, which includes effects of within-group and among-group variance and can be calculated from data. However, the quantity $\sigma^2_{Gt}$ can be estimated from the relationship

$$\sigma^2_{Gt} = \sigma^2_{Gw}/(1 - r_0), \tag{11}$$

where $\sigma^2_{Gw}$ is the pooled within-group genetic variance and $r_0$ is the weighted average genetic distance to the centroid. This relationship was first noted by Wright (1951) and was most recently confirmed by Rogers and Harpending (1983), who showed that it also holds under gametic disequilibrium. The pooled within-subdivision genetic variance should be weighted by census size and is calculated as

$$\sigma^2_{Gw} = \sum w_i \sigma^2_{Gi}, \tag{12}$$

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where $w_i$ is the relative census size of the $i$th subdivision and summation is over all subdivisions.

The parameter $r_0$ in Eq. (11) is given by

$$r_0 = \sum w_i r_{ii}. \tag{13}$$

Here, $r_0$ is equivalent to one of the two meanings of Wright's $F_{ST}$, which is also a measure of variation of distances to a regional centroid. One interpretation is the genetic distance to a centroid defined by allele frequencies in a founding population. Because such data are rarely available, especially for human populations, a second interpretation refers to variation about a centroid defined by contemporary allele frequencies. To avoid confusion, we follow Rogers and Harpending (1983, 1986) and use the symbol $F_{ST}$ to refer to the former interpretation and $r_0$ to refer to the latter. Note that $r_0$ is the same quantity designated as $R_{ST}$ by some researchers [e.g., Workman et al. (1973)].

The Harpending-Ward model can now be expressed as

$$E(\sigma^2_{Gi}) = \sigma^2_{Gw}(1 - r_{ii})/(1 - r_0). \tag{14}$$

To use this model, we must derive the genetic distances from the centroid ($r_{ii}$) from quantitative traits. Let $\bar{x}_t$ be the phenotypic mean of the entire region, defined as

$$\bar{x}_t = \sum w_i \bar{x}_i, \tag{15}$$

where $\bar{x}_i$ is the phenotypic mean for subdivision $i$ and summation is over all subdivisions. We assume that environmental deviations have expectation zero, so $\bar{x}_i$ also reflects the genetic mean of the trait. For $g$ groups, let $C$ be a $g \times g$ matrix with elements

$$c_{ij} = (\bar{x}_i - \bar{x}_t)(\bar{x}_j - \bar{x}_t). \tag{16}$$

Given the relationship between allele frequency and a phenotypic mean under an equal and additive effects model ($\bar{x}_i = 2\alpha p_i$) (Falconer 1981), the elements of the relationship matrix $R$ can now be written as

$$r_{ij} = c_{ij}(1 - r_0)/2\sigma^2_{Gw}. \tag{17}$$

The genetic distance of subdivision $i$ to the regional centroid is given by

$$r_{ii} = c_{ii}/(2\sigma^2_{Gw} + \sum w_i c_{ii}), \tag{18}$$

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where summation is over all subdivisions. The weighted average genetic distance to the centroid is now

$$r_0 = \sum w_i c_{ii} / (2\sigma^2_{Gw} + \sum w_i c_{ii}). \tag{19}$$

We do not consider sampling error in $r_0$ or the $r_{ii}$ values. If desired, these estimates could be adjusted by a suitable bias correction, as Workman et al. (1973) do.

The model is not dependent on the assumption of the simple genetic model in which two alleles exist at each locus. We made this assumption purely to show the relationship with the Harpending-Ward approach, which uses heterozygosity. The model can be derived directly from Eq. (14), which was first obtained by Wright (1951) and is appropriate for other forms of allelic distributions.

The practical requirements for using the method are stringent. Genetic variances are required for each subdivision. Possession of such information is rare. Generally, if any information on the additive genetic variance of a trait is available, it usually reflects a single subdivision or a pooled estimate. Therefore it is useful to adjust the method to predict phenotypic variances. Phenotypic variances for each subdivision are available in most situations. However, translation of the results to phenotypic variances also requires more restrictive assumptions.

Let the genetic variance of a trait in the $i$th subdivision be written as $\sigma^2_{Gi} = h^2_i \sigma^2_{Pi}$, where $\sigma^2_{Pi}$ is the phenotypic variance. Given the additional assumption that heritability is the same over all populations ($h^2_i = h^2$ for all $i$), the additive genetic variance of the panmictic region is $\sigma^2_{Gt} = h^2 \sigma^2_{Pt}$. The assumption of equal heritabilities over all subdivisions in a region is probably not too serious, given the small study areas of most anthropological investigations.

Using this scheme, direct substitution into Eq. (14) yields

$$E(\sigma^2_{Pi}) = \sigma^2_{Pw}(1 - r_{ii})/(1 - r_0). \tag{20}$$

The observed phenotypic variance in subdivision $i$ ($\sigma^2_{Pi}$) can be estimated as the sample variance. The $R$ matrix can now be written as

$$r_{ij} = c_{ij}(1 - r_0)/2h^2\sigma^2_{Pw}. \tag{21}$$

This equation is similar to that used by Morton et al. (1971), except that they took the total phenotypic variance as equivalent to the phenotypic variance expected under panmixia. As shown, the correct method utilizes the relationship between within-group phenotypic variance and the phenotypic variance expected under panmixia.

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The univariate method can be used with multiple traits by taking the average of $\sigma^2_{Pi}, \sigma^2_{Pw}, r_{ii}$, and $r_0$ over all traits. To average results over multiple traits, one must convert the sample means and variances into standardized scores. The univariate method has the advantage of being applicable using only summary statistics (means and phenotypic standard deviations) and the census size of each population (to compute the weighting coefficient $w_i$). However, to best account for intercorrelation of variables, a multivariate approach is preferred.

## Multivariate Extension to Quantitative Traits

The multivariate extension considers $g$ groups and $m$ variables. In addition, the assumptions required are necessarily more complex than in the univariate case. Let $P_i$ denote the phenotypic covariance matrix of the $i$th subdivision. We assume that $P_i$ can be decomposed into two constituent matrices: the additive genetic covariance matrix $G_i$ and the random environmental covariance matrix $E_i$. The multivariate generalization of Eq. (14) is given by

$$E(G_i) = G_w(1 - r_{ii})/(1 - r_0), \tag{22}$$

where $G_w$ is the pooled within-subdivision additive genetic covariance matrix. Equation (22) implies that all individual within-subdivision genetic covariance matrices are proportional to $G_w$, which in turn implies that there is a common genetic correlation matrix ($R_G$).

We use these requirements to derive a single index that parameterizes multivariate variability. The proportionality of the within-subdivision genetic covariance matrices implies that a common orthogonalization exists (i.e., that the matrices share the same eigenvectors). Therefore we can rewrite Eq. (22) using the decomposition

$$E(SL_{Gi}S') = SL_{Gw}S'(1 - r_{ii})/(1 - r_0), \tag{23}$$

where $S$ is the matrix of common eigenvectors, $L_{Gi}$ denotes a diagonal matrix containing the eigenvalues ($\lambda_{Gi(k)}$, which refers to the $k$th eigenvalue) of the $i$th subdivision's genetic covariance matrix, and $L_{Gw}$ contains the eigenvalues of the pooled genetic covariance matrix. From E

$$E(L_{Gi}) = L_{Gw}(1 - r_{ii})/(1 - r_0), \tag{24a}$$

$$E(\lambda_{Gi(k)}) = \lambda_{Gw(k)}(1 - r_{ii})/(1 - r_0), \tag{24b}$$

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$$E(\bar{\lambda}_{Gi}) = \bar{\lambda}_{Gw}(1 - r_{ii})/(1 - r_0), \tag{24c}$$

where $\bar{\lambda}$ denotes the mean eigenvalue of the matrix in question. This index, $\bar{\lambda}_{Gi}$, is convenient because it can be calculated as Tr $(G_i)/m$, where Tr$(\cdot)$ denotes the trace of a matrix and $m$ is the number of traits. It can be interpreted as the mean genetic variance over all $m$ traits, which we write simply as $\bar{v}_{Gi}$. The predicted mean genetic variance is then

$$E(\bar{v}_{Gi}) = \bar{v}_{Gw}(1 - r_{ii})/(1 - r_0). \tag{25}$$

Like the univariate case, predictions can be made at the level of the phenotypic variance using phenotypic covariance matrices in Eqs. (22)–(25). However, additional assumptions have to be made. Prediction of mean phenotypic variances requires proportionality of all phenotypic covariance matrices with all genetic covariance matrices. Such proportionality can be achieved only if all phenotypic and genetic covariance matrices share the same correlation structure and if there is a single common heritability for all traits and all subdivisions.

Multivariate estimation of the $R$ matrix follows the same logic as in the univariate case. Let $X$ be a $g \times m$ matrix containing the means of the subdivisions. A matrix of deviations ($\Delta$) from the population centroid can be written as

$$\Delta = (I - \mathbf{1}\mathbf{W}')X, \tag{26}$$

where $I$ is a $g \times g$ identity matrix, $\mathbf{1}$ is a vector of $g$ ones, and $\mathbf{W}$ is the relative census size vector. The $C$ matrix is now defined as

$$C = \Delta G_w^{-1} \Delta', \tag{27}$$

where $G_w^{-1}$ is the inverse of the pooled within-subdivision additive genetic covariance matrix and the prime denotes the transpose of the matrix. The $R$ matrix is then written as a function of $C$:

$$R = C(1 - r_0)/2m. \tag{28}$$

This equation differs from the univariate case only in its division by the number of traits ($m$) [see also Williams-Blangero and Blangero (1990)].

The genetic distance from the $i$th subdivision to the regional centroid can now be calculated as

$$r_{ii} = c_{ii}/(2m + \sum w_i c_{ii}), \tag{29}$$

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and the average genetic distance is

$$r_0 = (\sum w_i c_{ii})/(2m + \sum w_i c_{ii}). \tag{30}$$

For Eqs. (29) and (30) summation is over all $g$ subdivisions. These quantities can then be inserted into Eq. (25).

Utilization of this multivariate method requires at least an estimate of the pooled within-subdivision additive genetic covariance matrix ($G_w$). This matrix can be estimated using traditional quantitative genetic methods, provided that sufficient data exist on family relationships. Because of the large data requirements for accurate estimation of $G$ matrices, actual estimates of the $G_i$ will rarely be available. We suggest several approximations for obtaining the $G_i$.

One method of obtaining the $G_i$ given an estimate of $G_w$ is to assume that the environmental covariance matrix ($E$) is constant for all subdivisions ($E_i = E_w$ for all $i$). This is probably the least restrictive assumption and is likely to hold for neutral polygenic traits. Given this assumption, the $G_i$ can be obtained as

$$G_i = P_i - E_w. \tag{31}$$

One limitation of Eq. (31) is that the estimated $G_i$ may not be positive-definite, although estimation of the $P_i$ could be constrained, given $E_w$, to meet this requirement.

In some cases an estimate of $G_w$ will not be available but estimates of the heritabilities will be. In this situation $G_i$ can be estimated by assuming that the genetic correlation matrix is equal to the phenotypic correlation matrix ($R_{Gi} = R_{Pi}$) and that heritabilities are common to each subdivision. Cheverud (1988) notes the high correspondence between genetic and phenotypic correlation matrices in general, suggesting that this approach may not be unrealistic. Given these assumptions, the $G_i$ are obtained by

$$G_i = H^{1/2} P_i H^{1/2}, \tag{32}$$

where $H$ is an $m \times m$ diagonal matrix containing the heritabilities. If suitable heritability estimates are not available, the $G_i$ matrices can be estimated using a reasonable single estimate of average heritability, $h^2$, as suggested by Cheverud (1988). That is, $G_i = h^2 P_i$.

When no knowledge of genetic variation exists, there are two other methods of analysis. The model can be applied with a wide range of heritabilities. However, we prefer a simpler approach in which the phenotypic variation is assumed to be solely due to genetic variation,

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which yields $G_i = P_i$. In this case estimated distances from the regional centroid ($r_{ii}$) are known to reflect minimum genetic distances (Williams-Blangero and Blangero 1990). In any case, our experience has been that the same pattern of deviations from the expected regression line result over a wide range, provided that heritabilities are not too low (usually less than 0.2). Choice of heritability does not affect the quantity $(1-r_{ii})/(1-r_0)$ to any great extent.

## Applications

We illustrate the utility of our models by applying them to two different data sets. In both applications independent historical or migration data are available to assess the usefulness of the models for detecting differential gene flow. These two data sets illustrate the different types of samples most often available in studies of human population structure. For one data set the traits are known to be selectively neutral [dermatoglyphics (Blangero 1988)], the additive genetic covariance matrix and heritabilities can be estimated, and the ratio of sample size to census size is fairly high. For the other data set the traits can have strong environmental and developmental correlates (anthropometrics), no information is available on genealogical relationships from which to estimate the additive genetic covariance matrix or heritabilities, and the ratio of sample size to census size is low.

The dermatoglyphic data set was collected in 1985 from the Jirel tribe in the Jiri region of eastern Nepal. The Jirels are subsistence farmers who live in small, relatively endogamous villages. The Jirels live in nine villages, all of which were sampled. Historically, the Jirels have been considered a hybrid group from two nearby populations, the Sunwars and the Sherpa (Blangero 1987; Williams-Blangero 1987). Approximately 29% of the total population was sampled.

Ulnar and radial ridge counts were taken for all digits of both left and right hands for 1015 children and adults (536 males and 479 females). These data were then adjusted for sex differences. Given the large number of variables (20), principal components analysis was used for data reduction. Six principal components were extracted. The first principal component (PC) is a size component, illustrating high correlations between the number of ulnar and radial ridge counts of all digits. The other five principal components represent different contrasts between ridge counts on different digits. A pooled additive genetic covariance matrix was estimated from pedigree relationships using a maximum likelihood method (Blangero 1987). These data are for 685 individuals in 185 nuclear fam-

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*Table 1.* **Jirel Populations: Sample and Census Sizes, Genetic Distances to the Centroid ($r_{ii}$), Observed Mean Genetic Variances ($\bar{v}_{Gi}$), Expected Mean Genetic Variances [$E(\bar{v}_{Gi})$], and Residual Variances**

| *Village* | *Sample Size* | *Census Size* | $r_{ii}$ | $\bar{v}_{Gi}$ | $E(\bar{v}_{Gi})$ | *Residual* $[\bar{v}_{Gi} - E(\bar{v}_{Gi})]$ |
|---|---|---|---|---|---|---|
| Upper Sikri | 36 | 229 | 0.0176 | 0.616 | 0.610 | 0.005 |
| Lower Sikri | 85 | 461 | 0.0066 | 0.656 | 0.617 | 0.039 |
| Kharayoban | 56 | 228 | 0.0419 | 0.773 | 0.595 | 0.178 |
| Ratomate | 67 | 447 | 0.0174 | 0.682 | 0.610 | 0.072 |
| Dhunge | 46 | 365 | 0.0135 | 0.666 | 0.613 | 0.053 |
| Bhandar | 41 | 420 | 0.0081 | 0.491 | 0.616 | −0.125 |
| Kot | 113 | 539 | 0.0038 | 0.560 | 0.619 | −0.059 |
| Jungu | 34 | 467 | 0.0393 | 0.517 | 0.597 | −0.080 |
| Chetrapa | 25 | 369 | 0.0223 | 0.634 | 0.607 | 0.027 |

$r_0 = 0.0174$; $\bar{v}_{Gw} = 0.610$.

ilies. The diagonal elements of the matrix, which are the heritabilities, are all moderate to high ($PC_1 = 0.938, PC_2 = 0.560, PC_3 = 0.644, PC_4 = 0.598, PC_5 = 0.583, PC_6 = 0.420$).

We restricted subsequent analyses of genetic structure to 503 adult individuals (age 21 years and older) to avoid complications resulting from age structure. The sample sizes and census population sizes for the nine Jirel villages are reported in Table 1. We estimated within-subdivision additive genetic covariances using Eq. (31), assuming a common environmental covariance matrix. Using our multivariate method, we found the average dispersion of populations around the centroid to be $r_0 = 0.0174$. The results of the multivariate extension of the Harpending-Ward model are reported in Table 1 and Figure 1. The most noticeable outlier to the expected regression line is the village of Kharayoban, which exhibits greater phenotypic variation than expected, suggesting greater than average admixture from outside the local region. Blangero (1987) found that Kharayoban was most similar to Sherpa populations, and this village's high variation may reflect greater Sherpa admixture. The village of Bhandar is also an outlier, showing less phenotypic variation than expected. This village was also found to have the lowest rate of endemicity of all nine villages (Blangero 1987). That is, Bhandar has had the most migration from the other eight villages in the local region. The higher rate of local migration may have diminished any effects of external admixture in this group.

To investigate the effect of differences in heritability estimation on the Harpending-Ward plot, we took the pooled within-group phenotypic

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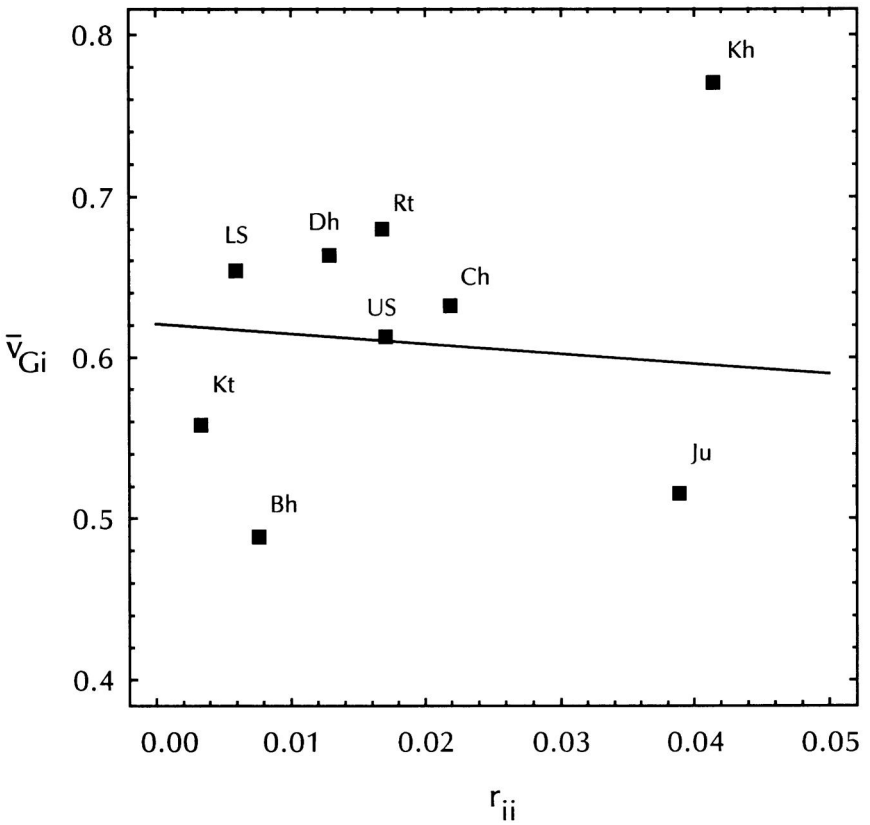



*Figure 1.* Plot of observed mean genetic variance ($\bar{v}_{Gi}$) versus genetic distance from the centroid ($r_{ii}$) for Jirel villages. The line indicates the expected regression line derived from the multivariate extension of the Harpending-Ward model. The $r_{ii}$ values were derived using the observed additive genetic covariance matrix $G$. US, Upper Sikri; LS, Lower Sikri; Kh, Kharayoban; Rt, Ratomate; Dh, Dhunge; Bh, Bhandar; Kt, Kot; Ju, Jungu; Ch, Chetrapa.

covariance matrix to estimate the additive genetic covariance using four estimates of average heritability ($h^2 = 1.00, 0.75, 0.50,$ and $0.25$). We then predicted within-subdivision mean phenotypic variances. The residual mean variances (observed mean variance − expected mean variance) are reported for all nine Jirel villages in Table 2. The residual variances obtained from direct use of the additive genetic covariance matrices are also shown for comparison. The residual variances are roughly the same regardless of the average heritability. The overall pattern of deviation for phenotypic variances is the same as that obtained from direct use of the additive genetic covariance matrix. Only if the heritabilities had

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*Table 2.* **Residual Mean Phenotypic Variances for Jirel Populations Using Different Average Heritabilities ($h^2$)**

| | *Residual Mean Variance* | | | | | |
|---|---|---|---|---|---|---|
| | *Predicted Mean Genetic Variance* | *Predicted Mean Phenotypic Variance* | | | | |
| *Village* | *Using Observed G Matrices* | *Using Pooled G Matrix* | $h^2 = 1.00$ | $h^2 = 0.75$ | $h^2 = 0.50$ | $h^2 = 0.25$ |
| Upper Sikri | 0.005 | 0.005 | 0.004 | 0.004 | 0.003 | 0.001 |
| Lower Sikri | 0.039 | 0.035 | 0.042 | 0.041 | 0.039 | 0.031 |
| Kharayoban | 0.178 | 0.187 | 0.176 | 0.181 | 0.191 | 0.219 |
| Ratomate | 0.072 | 0.072 | 0.075 | 0.076 | 0.079 | 0.086 |
| Dhunge | 0.053 | 0.052 | 0.053 | 0.052 | 0.050 | 0.044 |
| Bhandar | −0.125 | −0.129 | −0.126 | −0.128 | −0.132 | −0.144 |
| Kot | −0.059 | −0.064 | −0.058 | −0.061 | −0.066 | −0.082 |
| Jungu | −0.080 | −0.071 | −0.084 | −0.081 | −0.076 | −0.058 |
| Chetrapa | 0.027 | 0.029 | 0.027 | 0.028 | 0.030 | 0.036 |

been extremely low would there have been any substantial difference in expected variance relative to those obtained from the additive genetic covariance matrix. Different values of heritability cause changes in both $r_{ii}$ and $r_0$ but have little effect on the quantity $(1-r_{ii})/(1-r_0)$ for moderate to high heritabilities. These results suggest that estimation of average heritability is not critical and that the methods are fairly robust.

Our second data set consists of anthropometric measurements for 259 adult males in 7 populations in western Ireland. These data were originally collected during the 1890s (Relethford 1988). Ten anthropometric measurements were analyzed following stepwise polynomial regression to remove age effects (stature; forearm length; hand length; head length, breadth, and height; nose length and breadth; and bizygomatic and bigonial breadth). The sample sizes and census sizes of the seven populations are listed in Table 3. The Aran Islands and Inishbofin are islands off the west coast of Ireland. Previous research has shown that these islands have experienced considerable English admixture as the result of soldiers coming to live on the island during the sixteenth and seventeenth centuries (Relethford 1983, 1988). The populations of Carna, Garumna, and Lettermullen are on or near the coast of County Galway and are geographically close to the Aran Islands. As such, these populations may have experienced a lesser degree of English admixture resulting from continued exchange of genes with the Aran Islands. The populations of

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*Table 3.* **Irish Populations: Sample and Census Sizes, Genetic Distances to the Centroid ($r_{ii}$), Observed Mean Variances ($\bar{v}_i$), Expected Mean Variances [$E(\bar{v}_i)$], and Residual Variances Computed Using an Average Heritability of $h^2 = 1$**

| *Village* | *Sample Size* | *Census Size* | $r_{ii}$ | $\bar{v}_i$ | $E(\bar{v}_i)$ | *Residual* $[\bar{v}_i - E(\bar{v}_i)]$ |
|---|---|---|---|---|---|---|
| Aran Islands | 23 | 2907 | 0.0950 | 1.025 | 0.894 | 0.131 |
| Inishbofin | 34 | 874 | 0.1144 | 0.956 | 0.875 | 0.081 |
| Carna | 37 | 1670 | 0.0574 | 1.049 | 0.931 | 0.118 |
| Garumna | 30 | 1706 | 0.0549 | 0.803 | 0.934 | −0.131 |
| Lettermullen | 32 | 769 | 0.0716 | 0.931 | 0.917 | 0.014 |
| Ballycroy | 45 | 2036 | 0.0447 | 0.958 | 0.944 | 0.014 |
| Erris | 58 | 3757 | 0.0096 | 0.854 | 0.978 | −0.124 |

$r_0 = 0.0545$; $\bar{v}_w = 0.934$.
Census sizes are for the year 1891 and are taken from Relethford (1988) with two exceptions: (1) the census size of Erris is 3757 and not 3759 as reported there, and (2) the census size for Carna is larger in the present paper because it includes several small communities that are represented in the anthropometric sample. Also note that Erris is misspelled in the earlier paper (as "Ennis").

Ballycroy and Erris are further north on the coast and would be expected to show less English admixture.

Because heritabilities are not available for the Irish populations, we took the conservative approach and performed the analysis with $h^2 = 1$ for all traits (other values were also used and found to have minimal effect on the pattern of residuals from the expected regression line). Using our multivariate method, we obtained an estimate of $r_0 = 0.0545$. This is the minimum possible $r_0$ value (Williams-Blangero and Blangero 1990). This estimate seems rather high, but this is due in part to the inclusion of both admixed and nonadmixed populations. When the two highly admixed populations (Aran Islands and Inishbofin) are removed from the analysis, the value of $r_0$ drops to 0.0277.

The results of the multivariate extension of the Harpending-Ward model are reported in Table 3 and Figure 2. Both the Aran Islands and Inishbofin are high positive outliers to the expected regression line, as expected given previous evidence of English admixture in these groups. Of the three populations located near the Aran Islands, the position of Carna suggests greater than average admixture, the position of Lettermullen suggests average admixture, and the position of Garumna suggests less than average admixture. Of the two geographically distant populations, the position of Ballycroy suggests average admixture and the position of Erris suggests less than average admixture. Although this variation about the expected regression line may appear somewhat confusing, the aver-

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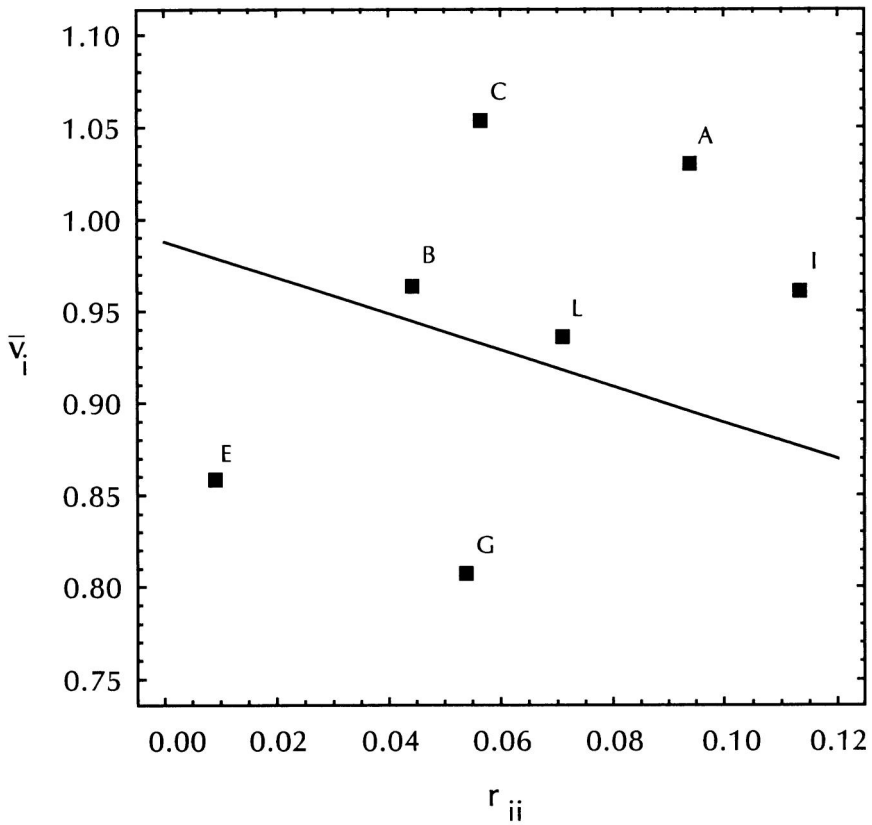



*Figure 2.* Plot of observed mean phenotypic (genetic) variance ($\bar{v}_i$) versus genetic distance from the centroid ($r_{ii}$) for Irish populations. The line indicates the expected regression line derived from the multivariate extension of the Harpending-Ward model. The $r_{ii}$ values were derived using a heritability of 1.0. A, Aran Islands; I, Inishbofin; C, Carna; G, Garumna; L, Lettermullen; B, Ballycroy; E, Erris.

age position of each of the three population clusters is quite revealing. The average residual mean variance (observed – expected) for the Aran Islands and Inishbofin is 0.106. The average for the three geographically close populations (Carna, Garumna, Lettermullen) is 0.001. The average for the distant and presumably least admixed populations (Ballycroy and Erris) is −0.055. Thus our model makes sense in terms of the known history of the region.

Although the average patterns of residual variation conform to our expectations, there is still considerable variation about the expected re-

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*Table 4.* **Irish Geographic Clusters: Genetic Distances to the Centroid ($r_{ii}$), Observed Mean Variances ($\bar{v}_i$), Expected Mean Variances [$E(\bar{v}_i)$], and Residual Variances Computed Using an Average Heritability of $h^2 = 1$**

| *Geographic Cluster* | $r_{ii}$ | $\bar{v}_i$ | $E(\bar{v}_i)$ | *Residual* $[\bar{v}_i - E(\bar{v}_i)]$ |
|---|---|---|---|---|
| Aran Islands | 0.0947 | 1.024 | 0.899 | 0.125 |
| Inishbofin | 0.1178 | 0.955 | 0.876 | 0.079 |
| County Galway | 0.0423 | 0.951 | 0.951 | 0.000 |
| County Mayo | 0.0137 | 0.904 | 0.979 | −0.075 |

$r_0 = 0.0462$; $\bar{v}_w = 0.947$.

gression line. For example, the three populations in County Galway (Carna, Garumna, and Lettermullen) show an *average* residual variance of approximately zero, but the range in residual variation among these three populations is high. Differential admixture may be a factor here, but the close geographic proximity of these populations argues against it. A more likely answer is variation resulting from small sample size. We repeated our analyses, grouping the seven populations into four geographic clusters. The Aran Islands and Inishbofin were kept as separate samples because of their geographic isolation and their overall phenotypic difference from each other and the other five populations (Relethford 1988). The populations of Carna, Garumna, and Lettermullen were pooled to form a cluster from County Galway. The populations of Ballycroy and Erris were pooled to form a cluster from County Mayo. This pooling is justifiable on the basis of close geographic proximity because previous canonical variates analysis has produced these clusters (Relethford 1988).

The results of our analysis of the four geographic clusters are reported in Table 4 and Figure 3. The admixed groups (Aran Islands and Inishbofin) lie above the expected regression line, as expected. The County Galway sample lies directly on the expected regression line, and the County Mayo sample lies below the expected regression line. Based on our knowledge of historical patterns of English admixture and geographic proximity, these results make perfect sense. These results also show that pooling geographically close and phenotypically similar samples may be appropriate in situations where sample size is relatively low.

## Discussion

To date, several studies have successfully used the Harpending-Ward model with allele frequency data to reveal differential external gene flow into a region. Our extensions to quantitative data also provide

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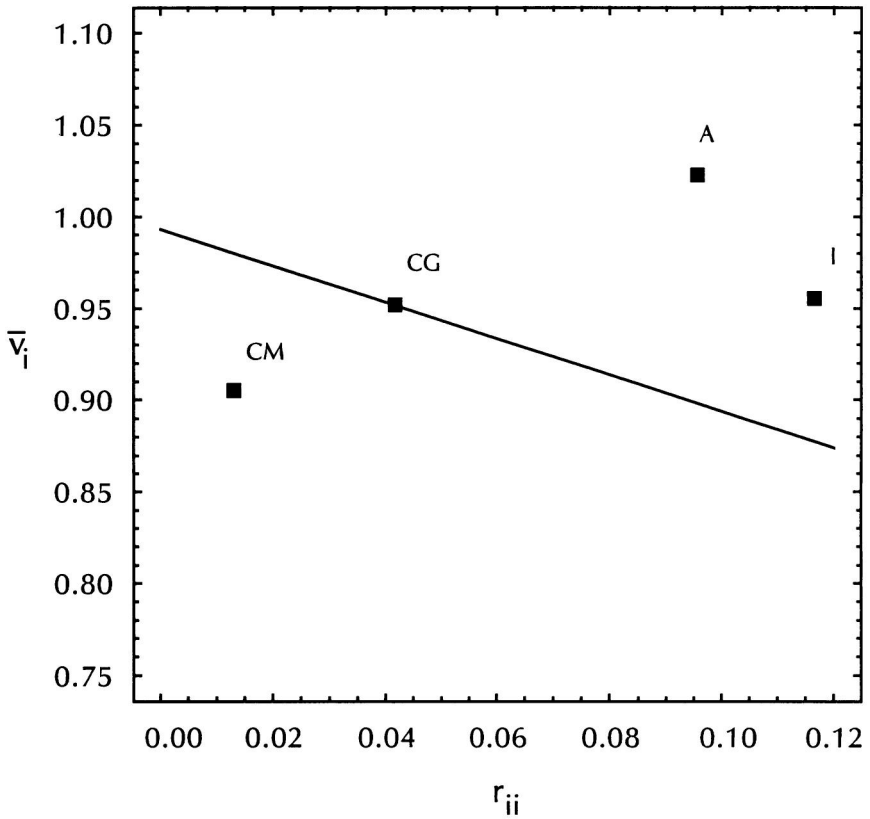



*Figure 3.* Plot of observed mean phenotypic (genetic) variance ($\bar{v}_i$) versus genetic distance from the centroid ($r_{ii}$) for Irish geographic clusters. The line indicates the expected regression line derived from the multivariate extension of the Harpending-Ward model. The $r_{ii}$ values were derived using a heritability of 1.0. A, Aran Islands cluster; I, Inishbofin cluster; CG, County Galway cluster; CM, County Mayo cluster.

results that are consistent with historical knowledge of the patterns of external migration. Development of these methods has also provided new estimates of $R$ matrix statistics from quantitative traits.

Our methods make a number of assumptions that must be kept in mind in any actual application. As in any model of genetic structure, the traits used are assumed to be selectively neutral. Given adequate independent data (e.g., allele frequencies for neutral loci or migration matrices), this assumption can be tested [e.g., Rogers and Harpending (1983)]. Another approach might be to repeat the analyses, deleting one

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variable at a time to detect possible common and unique patterns of among-group variation.

To some extent, the appropriateness of the assumptions can be addressed by judging the overall results. When independent historical information is available, such as in our applications here, the utility of the methods can be judged by their correspondence to expected patterns. In other cases comparisons can be made with other sources of data, such as allele frequencies. Such a comparative approach continues to be one of the best techniques for investigating genetic structure.

Any model of genetic structure that is applied to quantitative traits requires knowledge of either the additive genetic covariance matrix or the heritabilities of individual traits. Such information is needed, for example, to derive estimates of the distance from the centroid ($r_{ii}$) or the average distance to the centroid ($r_0$). In terms of our extensions to the Harpending-Ward model, however, the exact choice of such measures is not critical. As shown here, the same pattern of results is obtained with a wide range of estimates. In particular, we note the small effect of substituting different average heritabilities for the Jirel data even when these estimates differ greatly from the actual known estimates. We suggest, therefore, that our methods are quite robust to imprecise knowledge of the underlying patterns of genetic inheritance. Again, it should be noted that our model assumes neutrality. Variables known or suspected to be nonneutral could be eliminated and analyses performed again (much like the standard methods for genetic markers).

Our methods could be extended further by using tests of significance for determining outliers to the expected regression line. However, such approaches may not be that appropriate given the small number of subpopulations in most anthropological studies. We prefer interpretation to be made in light of independent information on history and migration and view our methods as heuristic devices to aid in an overall study of genetic structure.

Future analyses using our methods will benefit from the comparison of results obtained from quantitative and single-locus traits. Such comparisons could be useful in addressing issues regarding the relative impact of microevolutionary forces on single and multiple locus traits. Although some earlier studies have suggested that quantitative traits are less affected by genetic drift and show patterns of gene flow to a greater extent than single-locus traits, theoretical work has suggested that single-locus and multiple-locus traits provide equal amounts of information on genetic structure (Rogers and Harpending 1983). Repeated application of our methods may be useful in determining under what conditions such expectations hold.

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Our methods may also have some impact on applications of population genetics models to macroevolutionary studies. Although it is doubtful that human paleontological data would ever be sufficient for direct application of our methods, inferences regarding the relative impact of gene flow can still be made with regard to macroevolutionary hypotheses. For example, Stringer and Andrews (1988) discuss alternative models of the origin of anatomically modern *Homo sapiens* with reference to predictions regarding among-group and within-group variation. What is not clear from their models is the effect of variation in gene flow (both in space and time) and how this might affect within-group variation. An understanding of the Harpending-Ward model and an idea of the processes leading to deviations from the model will be useful in refining such ideas further.

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